

Term Deposit Prediction: Report

Overview

This project builds a predictive system to determine whether a client will subscribe to a term deposit. The approach combines exploratory data analysis (EDA), machine learning, and API deployment.

Dataset

- File Used: bank-additional-full.csv
- Target Variable: y (binary: 'yes' or 'no')

Methodology

- Categorical variables were one-hot encoded.
- SMOTE was applied to balance the dataset.
- Random Forest Classifier was selected for its robustness.
- Feature names were saved to ensure consistent preprocessing at inference time.

Model Performance

- Accuracy: 95.96%
- Precision: 95.91%
- Recall: 96.01%
- F1 Score: 95.96%
- ROC AUC Score: 95.96%

Findings and Insights

- Clients with longer lasting contact durations tend to subscribe.
- Previous campaign outcomes (especially success) greatly influence future engagement.
- Campaigns in certain months perform poorly (e.g., May).
- Over-contacting clients decreases conversion chances.

Key Predictors: duration, poutcome, contact, month, emp.var.rate

Deployment

- Backend: Flask API deployed on Render
- Frontend: HTML form hosted on Vercel, calls the Render API